QTL Mapping of Cotton Fibre Quality Traits: A Comparison of Classical and Regularised Approaches

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Abstract

Identifying genomic loci associated with quantitative traits remains a central challenge in plant genetics, particularly in the high-dimensional regime where the number of markers substantially exceeds the number of genotyped individuals. This study analyses a cotton Multi-parent Advanced Generation Inter-Cross (MAGIC) population comprising 256 lines genotyped across 6,523 markers and phenotyped for four fibre quality traits — lint percentage, seed index, lint index, and seed oil content — over three field seasons. Four marker selection approaches are evaluated: a season-adjusted single-marker scan, a principal component (PC)-adjusted variant, lasso multi-marker regression, and AIC-guided stepwise forward selection. All four traits exhibit high narrow-sense heritability (genomic restricted maximum likelihood estimates ranging from $\widehat{h}^2 = 0.62$ to 0.83) estimated from a genomic mixed-effects model using a genomic relationship matrix (GRM) built from all 6,523 markers. Single-marker scans with Benjamini–Hochberg correction select large numbers of markers per trait, reflecting the high linkage disequilibrium inherent to MAGIC populations. Lasso regression with the one-standard-error rule (λ_{1se}) selects a compact set of markers and offers substantially improved precision at moderate cost in power. A factorial simulation study (3×3 design crossing genetic architecture — Sparse ($n_{\text{causal}} = 10$), Moderate ($n_{\text{causal}} = 25$), and Polygenic ($n_{\text{causal}} = 100$) — with signal strength $h_g^2 \in \{0.10, 0.20, 0.35\}$, season heritability fixed at $h_s^2 = 0.20$, 50 replicates per cell) uses the observed cotton genotype matrix to preserve the real LD structure. Results confirm that single-marker scans achieve substantially higher power than lasso but with false-discovery rate (FDR) near 0.98–0.99 across all conditions — a structural consequence of the high LD in this MAGIC panel — while lasso provides substantially improved precision and F1 score at moderate cost in power. The most pronounced advantage of lasso over the scan methods is observed under moderate signal ($h_g^2 = 0.20$) and Sparse-to-Moderate causal architectures. An interactive R Shiny companion application, integrated as a teaching and exploratory tool, allows users to replicate the simulation framework, adjust design parameters, and visualise the power-FDR trade-off in real time. Together, the empirical analysis and simulation study provide a coherent methodological framework for evaluating QTL mapping strategies in high-dimensional cotton breeding datasets.

Keywords: QTL mapping, MAGIC population, lasso regression, linkage disequilibrium, power analysis, Shiny application, *Gossypium hirsutum*

1 Introduction

Quantitative trait locus (QTL) mapping aims to identify genomic regions whose allelic variation contributes to phenotypic differences in complex, quantitatively varying traits. In the classical framework described by Haley and Knott (1992), individual markers are tested one at a time via regression of phenotype on genotype score. Broman and Speed (2002) later placed this single-marker approach within a broader framework that encompasses interval mapping and multiple-QTL model selection. These classical methods were designed for relatively sparse marker panels in structured crosses — backcross, F_2, or recombinant inbred line populations — where the number of markers p is modest relative to the number of genotyped individuals n.

Modern high-throughput genotyping platforms make this $p \ll n$ assumption untenable. Dense SNP arrays and genotyping-by-sequencing routinely generate marker panels in which p exceeds n by one or more orders of magnitude, creating what statisticians refer to as the large p, small n problem (Hastie, Tibshirani, and Friedman 2009). Under this regime, ordinary least-squares regression is underdetermined: no unique solution exists, and even approximate solutions are numerically unstable. Furthermore, the dense marker panels typical of modern breeding programmes exhibit strong linkage disequilibrium (LD) among physically proximate loci, so the assumption of independent tests that underlies classical Bonferroni or permutation-based multiple-testing corrections becomes overly conservative and reduces statistical power to detect true associations (Li et al. 2018).

Multi-parent Advanced Generation Inter-Cross (MAGIC) populations have been developed in several crop species specifically to address the resolution limitations of biparental crosses (Kover et al. 2009). By intercrossing multiple founder lines over several generations, MAGIC designs accumulate a richer history of recombination events, resulting in shorter LD blocks and finer mapping resolution than a standard F_2 or backcross population. However, the multi-founder structure also introduces allelic heterogeneity and complex population structure that must be properly accounted for in statistical models, either through explicit PC covariates (Price et al. 2006) or through genomic mixed-effects models that incorporate a genomic relationship matrix (VanRaden 2008; Yu et al. 2006).

This study analyses a cotton MAGIC population containing 256 genotyped lines and 6,523 markers — a ratio of $p/n \approx 25$ — phenotyped for four economically important fibre quality traits across three field seasons. The four traits are lint percentage, seed index, lint index, and seed oil content. These traits are of direct relevance to upland cotton (*Gossypium hirsutum*) breeding because they are key determinants of fibre yield and seed composition (Li et al. 2022). A strong negative phenotypic correlation between lint percentage and seed index is known to create a yield–quality tension in boll development, and QTL studies have identified genomic regions with antagonistic effects on both traits simultaneously.

Four marker selection approaches are compared. The first is a season-adjusted single-marker scan used as a classical baseline. The second adds genotype PCs as additional covariates to assess whether population structure influences the marker signals (Price et al. 2006). The third is lasso multi-marker regression, which jointly selects markers under an ℓ_1 sparsity penalty and is well suited to the $p \gg n$ regime (Tibshirani 1996; Friedman, Hastie, and Tibshirani 2010). The fourth is AIC-guided stepwise forward selection applied to a pre-filtered candidate pool, representing a compromise between the marginal approach of the scans and the full joint modelling of lasso (Akaike 1974). Heritability is estimated independently using a genomic mixed-effects model to provide context for interpreting the power of the marker selection methods.

To evaluate all four approaches under controlled conditions where the true causal marker set is known, a factorial simulation study is conducted using the observed cotton genotype matrix. The simulation crosses three genetic architecture levels ($n_{\text{causal}} \in \{10, 25, 100\}$) with three signal-strength levels ($h_g^2 \in \{0.10, 0.20, 0.35\}$), with 50 replicates per cell, yielding a comprehensive $3 \times 3 \times 50$ evaluation of power, precision, false-discovery rate, and F1 score.

An interactive R Shiny companion application, the QTL Design Sandbox, is developed as an integrated teaching and exploratory tool that mirrors the simulation framework described in Section 3.7. The application allows users to upload their own genotype data or use a built-in toy dataset, simulate quantitative traits under user-defined QTL parameters, fit single-marker or multi-marker models, and visualise the resulting power and FDR curves interactively. The design and statistical logic of the Shiny tool are described in Section 3.8, where its connections to the main analytical framework are made explicit.

The remainder of this paper is structured as follows. Section 2 describes the data and exploratory phenotypic patterns. Section 3 presents all four statistical methods together with the simulation design and the Shiny application. Section 4 reports the empirical marker selection results, heritability estimates, and simulation-based performance metrics. Section 5 interprets the findings, compares them to the existing cotton QTL literature, and describes the limitations and future directions of this work.

2 Data

2.1 Raw data and design structure

The dataset used in this study originates from a cotton MAGIC population described by Li et al. (2022). The genotype file contains SNP marker scores for 256 cotton lines across 6,523 marker positions, coded on the $\{-1, 0, 1\}$ scale representing the two homozygous classes and the heterozygous class respectively. The genetic map file provides the chromosomal position in centimorgans (cM) for each marker. The phenotype

Table 1. Descriptive statistics for the four analysed traits by field season and overall. Mean (SD) and Median [Min, Max] are reported.

Trait	Season 1 N = 256	Season 2 N = 256	Season 3 N = 256	Overall N = 768
Lint percentage				
Mean (SD)	39.50 (1.92)	43.91 (1.95)	43.93 (2.03)	42.45 (2.87)
Median [Min, Max]	39.50 [30.70, 44.90]	43.90 [34.00, 49.20]	43.90 [31.60, 49.40]	42.80 [30.70, 49.40]
Seed index				
Mean (SD)	9.58 (0.70)	8.15 (0.50)	7.73 (0.50)	8.49 (0.98)
Median [Min, Max]	9.50 [7.30, 12.30]	8.10 [7.00, 10.50]	7.65 [6.30, 10.50]	8.20 [6.30, 12.30]
Lint index				
Mean (SD)	7.54 (0.56)	7.62 (0.56)	7.66 (0.52)	7.61 (0.55)
Median [Min, Max]	7.50 [5.90, 9.30]	7.60 [6.10, 9.10]	7.60 [6.20, 9.40]	7.60 [5.90, 9.40]
Seed oil content				
Mean (SD)	19.07 (1.10)	18.38 (1.09)	21.68 (1.08)	19.71 (1.79)
Median [Min, Max]	19.10 [15.90, 24.00]	18.30 [15.20, 24.40]	21.60 [19.30, 26.90]	19.30 [15.20, 26.90]

file records measurements of four fibre quality traits — lint percentage, seed index, lint index, and seed oil content — for all 256 lines observed in three separate field seasons, yielding 768 phenotypic records in total.

Prior to analysis, missing marker values were replaced by the column mean across all 256 lines, so that no observations were lost due to sporadic missing genotype calls. All 6,523 markers have positive variance across lines, so none were removed. The phenotype file requires no imputation, as all 768 records are complete. The season indicator was reconstructed by assigning the first 256 rows to Season 1, the next 256 to Season 2, and the final 256 to Season 3, consistent with a balanced multi-environment trial structure. The merged analysis dataset therefore contains 768 rows (one per line–season combination) and 6,523 marker columns alongside the phenotype and design variables.

All analyses were conducted in R (R Core Team 2024) using the following packages: `glmnet` (Friedman, Hastie, and Tibshirani 2010) for lasso regression; `irlba` (Baglama and Reichel 2005) for truncated principal component analysis; `lme4breeding` (Covarrubias-Pazaran 2023) for the genomic mixed-effects model; `gtsummary` (Sjoberg et al. 2021) and `kableExtra` (Zhu 2024) for table formatting; and the `tidyverse` (Wickham et al. 2019) suite for data manipulation and visualisation.

2.2 Exploratory analysis of phenotypic traits

Table 1 reports descriptive statistics for each trait by season and overall. Lint percentage is substantially lower in Season 1 (mean 39.50) than in Seasons 2 and 3 (means 43.91 and 43.93 respectively). Seed index follows the opposite pattern, being highest in Season 1 (9.58) and declining in later seasons. Lint index is relatively stable across seasons with means between 7.54 and 7.66. Seed oil content is lowest in Season 2 (18.38) and highest in Season 3 (21.68). These mean shifts are large enough to confound marker–trait associations if season is not included as a covariate, which directly motivates its inclusion in all regression models described in Section 3.

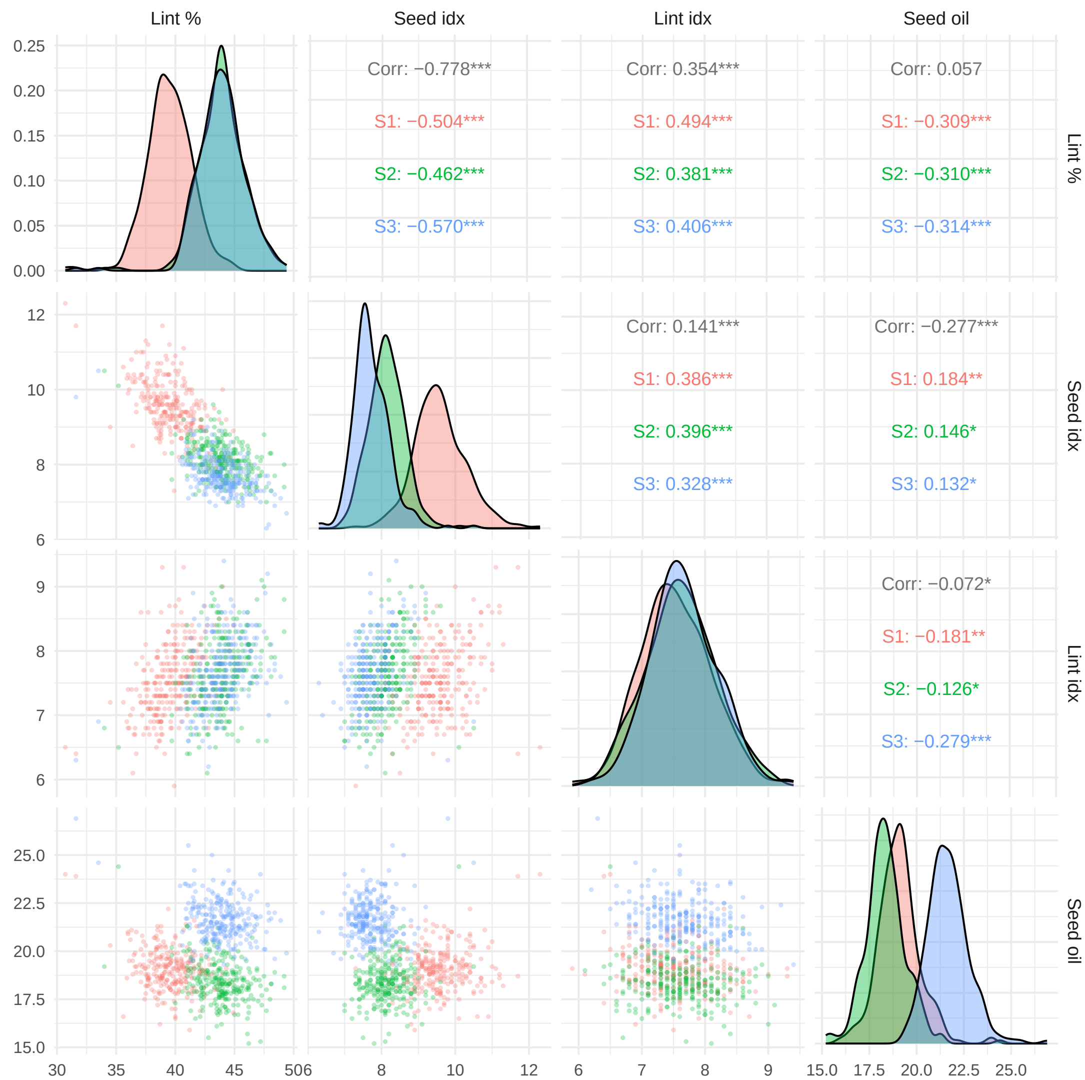

Figure 1. Pairwise scatterplot matrix for the four analysed traits, coloured by season. Lower-triangular panels show raw scatterplots. Diagonal panels show season-specific kernel density curves. Upper-triangular panels report the overall Pearson correlation and within-season correlations.

Figure 1 reveals a strong negative phenotypic correlation between lint percentage and seed index ($r = -0.778$ overall). This antagonistic relationship reflects a fundamental carbon source–sink trade-off in cotton boll development: assimilates directed toward lint fibre elongation are unavailable for seed filling. Lint percentage and lint index are positively correlated ($r \approx 0.62$), as both measure aspects of lint accumulation. Seed oil content is weakly correlated with the other three traits, consistent with its distinct metabolic pathway.

2.3 Genotype principal components and population structure

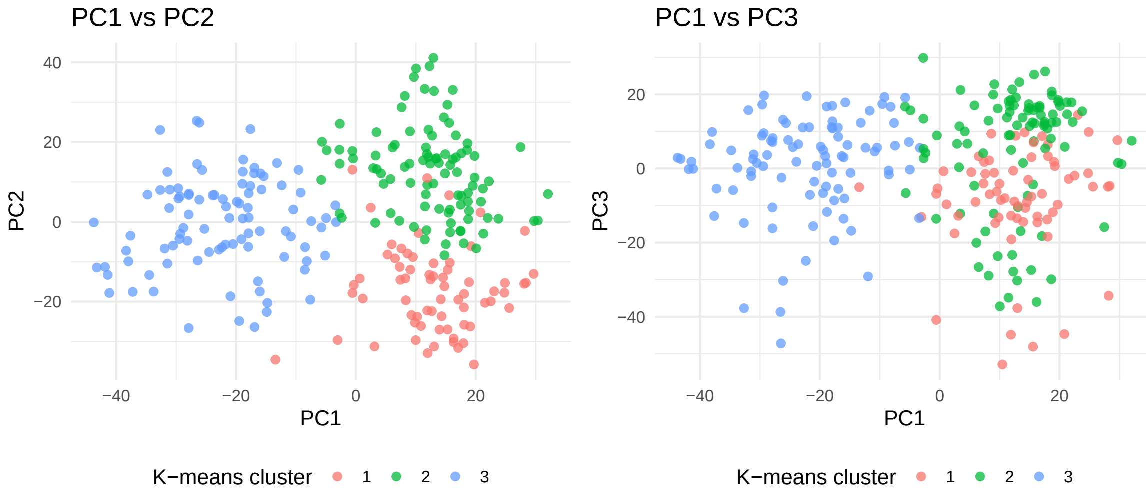

Figure 2. Scatterplots of the first three genotype principal components coloured by k-means cluster. The broad but continuous genetic structure suggests gradual rather than sharply discrete population stratification.

Figure 2 shows that the cotton lines form broad genetic groups with continuous rather than sharply separated boundaries. Population structure is present but gradual, suggesting that including the leading genotype PCs as covariates is sufficient to control for stratification without requiring a formal genomic random-effects correction in the single-marker scans.

3 Methods

3.1 Single-marker regression models

The most common approach to QTL mapping is single-marker regression, in which each marker is tested independently for association with the trait. For marker j and trait observation y_{is} (line i, season s), the additive model is:

$$y_{is} = \alpha + \gamma_s + \beta_j x_{ij} + \varepsilon_{is}, \qquad \varepsilon_{is} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \tag{1}$$

where α is the intercept, γ_s are season fixed effects, x_{ij} is the genotype score of line i at marker j coded as $\{-1, 0, 1\}$, and β_j is the additive marker effect. The null hypothesis H_0: $\beta_j = 0$ is tested using a Wald t-statistic with $(n - \text{rank}(\mathbf{C}) - 1)$ residual degrees of freedom, where $\mathbf{C}$ is the covariate design matrix. Computationally, all $p = 6{,}523$ markers are evaluated efficiently by QR decomposition of the covariate matrix, residualising both $\mathbf{y}$ and $\mathbf{X}$ with respect to the covariates, and computing the inner-product statistics on the residual matrices (Haley and Knott 1992).

A second scan specification adds the first five genotype principal components as additional covariates, replacing γ_s in equation Equation 1 with both season indicators and the PC scores:

$$y_{is} = \alpha + \gamma_s + \sum_{k=1}^{5} \delta_k \text{PC}_{ik} + \beta_j x_{ij} + \varepsilon_{is}. \tag{2}$$

Including the PCs adjusts for the broad genetic structure visible in Figure 2 and can attenuate associations that are driven by population stratification rather than local marker effects (Price et al. 2006).

3.2 Lasso multi-marker regression

In the high-dimensional setting where $p \gg n$, single-marker scans remain valid as marginal screening tools but cannot jointly estimate the effects of all markers simultaneously. The lasso (Tibshirani 1996) addresses this by estimating the full vector of marker effects β subject to an ℓ_1 penalty:

$$\min_{\alpha,\beta,\gamma,\delta} \frac{1}{n} \|\mathbf{y} - \alpha\mathbf{1} - \mathbf{G}\beta - \mathbf{C}_{\text{season}}\gamma - \mathbf{C}_{\text{PC}}\delta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j|, \tag{3}$$

where $\mathbf{G}$ is the $n \times p$ genotype matrix, $\mathbf{C}_{\text{season}}$ and $\mathbf{C}_{\text{PC}}$ are covariate matrices for season and PC effects, and $\lambda > 0$ is the penalty parameter. Crucially, only the marker-effect vector β is penalised; the season and PC coefficients γ and δ are included as unpenalised adjustment covariates. The ℓ_1 penalty induces sparsity by shrinking many $\hat{\beta}_j$ exactly to zero, retaining only the markers most informative for prediction after accounting for the correlation structure of the marker panel (Tibshirani 1996).

The penalty parameter λ is chosen by ten-fold cross-validation using the `cv.glmnet` function from the `glmnet` package (Friedman, Hastie, and Tibshirani 2010). Two solutions are considered: λ_{min}, which minimises cross-validated prediction error, and λ_{1se}, which is the largest λ whose error is within one standard error of the minimum. Because λ_{1se} applies a stronger penalty, it yields a smaller and more stable selected set; it is used as the primary reported result.

3.3 Mixed-effects model and heritability estimation

A genomic linear mixed-effects model is fitted to obtain trait-level heritability estimates independently of the marker selection procedure. For line i observed in season s:

$$y_{is} = \alpha + \gamma_s + u_i + \varepsilon_{is}, \quad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \sigma_g^2\mathbf{G}), \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \sigma_e^2\mathbf{I}), \tag{4}$$

where u_i is the random genetic effect for line i and $\mathbf{G}$ is the genomic relationship matrix (GRM). Because the SNP markers are coded on the $\{-1, 0, 1\}$ scale rather than the standard $\{0, 1, 2\}$ allele-dosage scale, a coding-agnostic GRM is constructed by first column-standardising the marker matrix $\mathbf{M}$ to mean zero and unit variance, and then computing:

$$\mathbf{G} = \frac{\mathbf{M}^*\mathbf{M}^{*\top}}{p}, \tag{5}$$

where $\mathbf{M}^*$ is the $n \times p$ standardised marker matrix. A small ridge term $(0.05 \times \mathbf{I})$ is added to the diagonal to guarantee positive definiteness. Variance components $\hat{\sigma}_g^2$ and $\hat{\sigma}_e^2$ are estimated by restricted maximum likelihood (REML), and narrow-sense heritability is computed as:

$$\hat{h}^2 = \frac{\hat{\sigma}_g^2}{\hat{\sigma}_g^2 + \hat{\sigma}_e^2}. \tag{6}$$

The model is fitted using the `lme4breeding` package (Covarrubias-Pazaran 2023) with season as a fixed effect using treatment (dummy) coding, with Season 1 as the reference level.

3.4 Stepwise multi-marker selection

The fourth approach applies AIC-guided stepwise forward-and-backward selection. Because fitting a full $p = 6{,}523$-marker model is computationally infeasible and numerically ill-conditioned, the candidate pool is first reduced to the 50 markers with the smallest marginal p-values from the PC-adjusted scan. Stepwise selection using the R `step()` function then iterates over this candidate pool, starting from the null model containing only the season and PC covariates, and adding or removing one marker at each step according to the AIC criterion. The final retained marker set is reported as the selected set for that trait. This approach differs from lasso in two key respects: it makes hard in/out decisions rather than applying continuous shrinkage, and it is guided by AIC rather than cross-validated prediction error. Its primary limitation relative to lasso is the risk that true causal markers may be discarded by the pre-filtering stage before stepwise selection begins.

3.5 Multiple-testing correction

Because the single-marker scans evaluate all 6,523 markers simultaneously, raw p-values require multiple-testing adjustment. Two methods are applied. The Benjamini–Hochberg (BH) procedure (Benjamini and Hochberg 1995) controls the expected false discovery rate (FDR) and is less conservative than the Bonferroni correction, which controls the family-wise error rate at threshold α/p. For the main comparisons, BH-significant sets are used because they provide a more practical balance between false-positive control and detection power in high-dimensional settings. Both correction results are reported for completeness.

3.6 Comparing selected marker sets

Pairwise overlap between selected marker sets is quantified using the Jaccard similarity index (Jaccard 1901):

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|},$$

which equals 1 when two sets are identical and 0 when they share no members. For the simulation study, let $\hat{\mathcal{S}}$ be the selected marker set and $\mathcal{C}$ the true causal set. Power (recall), precision, FDR, and the F1 score are defined as:

$$\text{Power} = \frac{|\hat{\mathcal{S}} \cap \mathcal{C}|}{|\mathcal{C}|}, \qquad \text{Precision} = \frac{|\hat{\mathcal{S}} \cap \mathcal{C}|}{|\hat{\mathcal{S}}|}, \qquad \text{FDR} = 1 - \text{Precision}, \qquad \text{F1} = \frac{2 \times \text{Precision} \times \text{Power}}{\text{Precision} + \text{Power}}.$$

The F1 score provides a single summary statistic that balances sensitivity and specificity and is used as the primary criterion for comparing methods in the simulation study.

3.7 Power analysis by simulation

To evaluate method performance under controlled conditions, phenotypes are simulated using the observed cotton genotype matrix so that the true marker correlation structure is preserved. This is essential because methods relying on marginal testing are affected by LD in ways that independence-based simulations would not capture.

For each replicate, n_{causal} markers are drawn uniformly at random. Their standardised genotype scores are combined into a genetic component $\mathbf{g} = \mathbf{X}_{\text{causal}}\beta_{\text{true}}$, where β_{true} is sampled from a standard normal distribution and normalised to unit ℓ_2 norm. A seasonal component and independent error are added:

$$y_{is} = \sqrt{h_g^2}\, g_i + \sqrt{h_s^2}\, s_i + \sqrt{1 - h_g^2 - h_s^2}\, \varepsilon_{is}, \tag{7}$$

where h_g^2 is the genetic heritability, $h_s^2 = 0.20$ is the season heritability, and all three components are scaled to unit variance before combining. All four methods are then fitted to the simulated phenotype and their selected marker sets are recorded. The simulation adopts a 3×3 factorial design crossing three genetic architecture levels ($n_{\text{causal}} \in \{10, 25, 100\}$, labelled Sparse, Moderate, and Polygenic) with three signal-strength levels ($h_g^2 \in \{0.35, 0.20, 0.10\}$), run for 50 replicates per cell.

Table 2. Factorial simulation design crossing genetic architecture (causal marker count) with signal strength (genetic heritability h_g^2). Season heritability $h_s^2 = 0.20$ is fixed throughout.

Architecture	Strength	n_{causal}	h_g^2	h_s^2	Error var.
Sparse (causal = 10)					
Sparse	Strong	10	0.35	0.2	0.45
Sparse	Moderate	10	0.20	0.2	0.60
Sparse	Weak	10	0.10	0.2	0.70
Moderate (causal = 25)					
Moderate	Strong	25	0.35	0.2	0.45
Moderate	Moderate	25	0.20	0.2	0.60
Moderate	Weak	25	0.10	0.2	0.70
Polygenic (causal = 100)					
Polygenic	Strong	100	0.35	0.2	0.45
Polygenic	Moderate	100	0.20	0.2	0.60
Polygenic	Weak	100	0.10	0.2	0.70

3.8 QTL Design Sandbox: interactive Shiny companion

An interactive R Shiny application, the **QTL Design Sandbox**, is developed as a companion tool to the analytical framework described in the preceding sections. The application makes the core statistical concepts of this study — trait simulation, marker selection modelling, power analysis, and FDR control — accessible to users without requiring them to write or execute R code directly. It is designed to serve two complementary purposes: as a teaching tool that illustrates the statistical principles behind each modelling approach, and as an exploratory sandbox that allows researchers to test design decisions before committing to a full analysis.

The application is structured around five functional modules that correspond directly to the analytical steps in this paper:

Data Input. Users may upload a custom CSV genotype matrix (individuals by markers) or load a built-in toy dataset. This module corresponds to the data described in Section 2: the expected input format is the same $n \times p$ genotype matrix used throughout the analysis.

Trait Simulation. Users specify the number of QTL (n_{causal}), the effect-size distribution, and the proportion of variance explained by genetics (h_g^2) and season (h_s^2). The module then generates a phenotype vector using the simulation model in Equation 7 applied to the uploaded or toy genotype matrix. This module mirrors Section 3.7 directly: by varying n_{causal} and h_g^2, users can explore the nine factorial cells of the simulation design in Table 2 interactively.

Model Fitting. Users choose between single-marker regression (corresponding to Section 3.1) or a multi-marker model (corresponding to Section 3.2), execute the chosen method, and view a results table reporting marker estimates, p-values, LOD scores, and likelihood-ratio test statistics. The visual output makes the contrast between marginal and joint testing immediately apparent: single-marker models flag broad genomic peaks, while multi-marker models retain only a few representative markers.

Power Analysis. This module implements the resampling-based framework from Section 3.7. Users specify a range of sample sizes and run either a bootstrap simulation or a λ/NCP-based theoretical power calculation. The output includes a power curve plot (proportion of true QTL detected vs. sample size) and an FDR curve plot (expected false-discovery proportion vs. sample size), directly visualising the trade-off that is analysed numerically in Section 4.5. The module also reports the minimum sample size required to achieve a user-specified target power, providing practical guidance for experimental design.

Output and Reporting. Summary tables, parameter settings, and plots — including power curves, FDR curves, and model result tables — can be exported as a self-contained HTML report. This feature supports reproducibility by documenting the exact simulation parameters and results used in any given session.

The Shiny application is available at [the project GitHub repository] and requires R $\geq$ 4.1 with the packages `shiny`, `ggplot2`, `DT`, `shinydashboard`, and `shinythemes` installed. An extended user guide is provided on the application's **Readme** tab. Together with the formal simulation results in Section 4.5, the Shiny tool provides a complete and reproducible framework for evaluating the power-FDR trade-off in QTL mapping under different design assumptions.

4 Results

4.1 Variance components and heritability estimates

Table 3. Variance component estimates from the genomic mixed-effects model (season fixed, genotype random with GRM). $\hat{\sigma}_g^2$: genetic variance; $\hat{\sigma}_e^2$: residual variance; $\hat{h}^2$: narrow-sense heritability.

Trait	$\hat{\sigma}_g^2$	$\hat{\sigma}_e^2$	$\hat{h}^2$	N
Lint percentage	4.518	0.948	0.826	768
Seed index	0.308	0.097	0.760	768
Lint index	0.334	0.068	0.831	768
Seed oil content	0.975	0.419	0.699	768

Table 3 presents REML variance component estimates for each of the four traits. All four traits show high narrow-sense heritability, with $\hat{h}^2$ ranging from 0.7 to 0.83. Lint percentage and lint index share the highest heritabilities, which is consistent with their strong and reproducible signals in subsequent marker scans. The observed $\hat{h}^2$ values substantially exceed the genetic heritability levels ($h_g^2 \in \{0.10, 0.20, 0.35\}$) used in the simulation study, implying that the simulation provides a conservative lower bound on the detectability of trait-associated markers in the real data.

4.2 Number of markers selected by each method

Table 4. Number of markers selected by each method for each trait. For the single-marker scans, results are shown under three multiple-testing correction thresholds: uncorrected ($p < 0.05$), Benjamini–Hochberg (BH) FDR at 5%, and Bonferroni at 5%. For lasso, the λ_{min} and λ_{1se} solutions are reported. For stepwise, the final AIC-selected set is shown.

Trait	Season scan (BH)	PC scan (BH)	Lasso PC (min)	Lasso PC (1se)	Lasso season (min)	Lasso season (1se)	Stepwise (AIC)
Lint percentage	1160	1740	240	160	231	165	18
Seed index	2121	1159	239	171	247	181	19
Lint index	1701	1711	222	190	220	171	18
Seed oil content	1389	942	212	140	215	161	14

Table 4 summarises the number of markers selected by each method. The season-only scan is the most liberal, selecting over 1,000 BH-significant markers for lint percentage and seed index. Adding PC covariates reduces the count for some traits (notably seed index and seed oil content) but increases it for others (lint index and lint percentage), showing that the relationship between population structure and individual trait signals is not uniform. The lasso selects far fewer markers in all cases: under λ_{1se}, the selected set ranges from 140 to 190 markers across the four traits, compared with several hundred to over two thousand for the scan methods. This contrast arises from the fundamentally different logic of the two approaches: marginal testing flags all markers with sufficiently small p-values, including many that are correlated with causal markers through linkage disequilibrium, while lasso imposes competition among markers and retains only the most informative representatives.

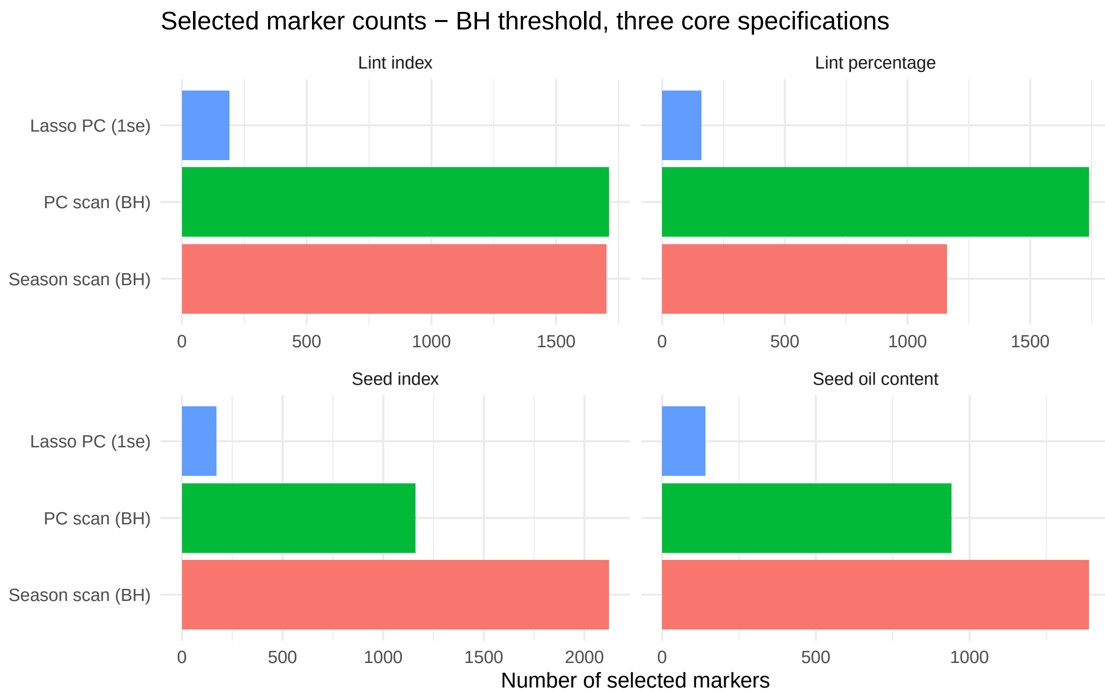

Figure 3. Number of BH-significant markers selected by the three core model specifications across the four traits. Methods are shown on the y-axis and marker counts on the x-axis to avoid label tilting. The large gap between the scan methods and lasso reflects their fundamentally different selection logic.

Figure 3 visualises the same information graphically, with methods on the y-axis so that labels require no tilting. The gap between scan-based methods and lasso is most pronounced for lint percentage and lint index, where both scans select over 1,000 markers while lasso retains fewer than 200 under λ_{1se}. The pattern is consistent across all four traits, confirming that the three methods are not simply producing the same answer with different thresholds — they address the marker selection problem in fundamentally different ways.

To visualise how much each pair of methods agrees on which markers to select, Figure 4 presents Venn diagrams for each trait using the BH-significant sets for the two scans and the λ_{1se} set for lasso.

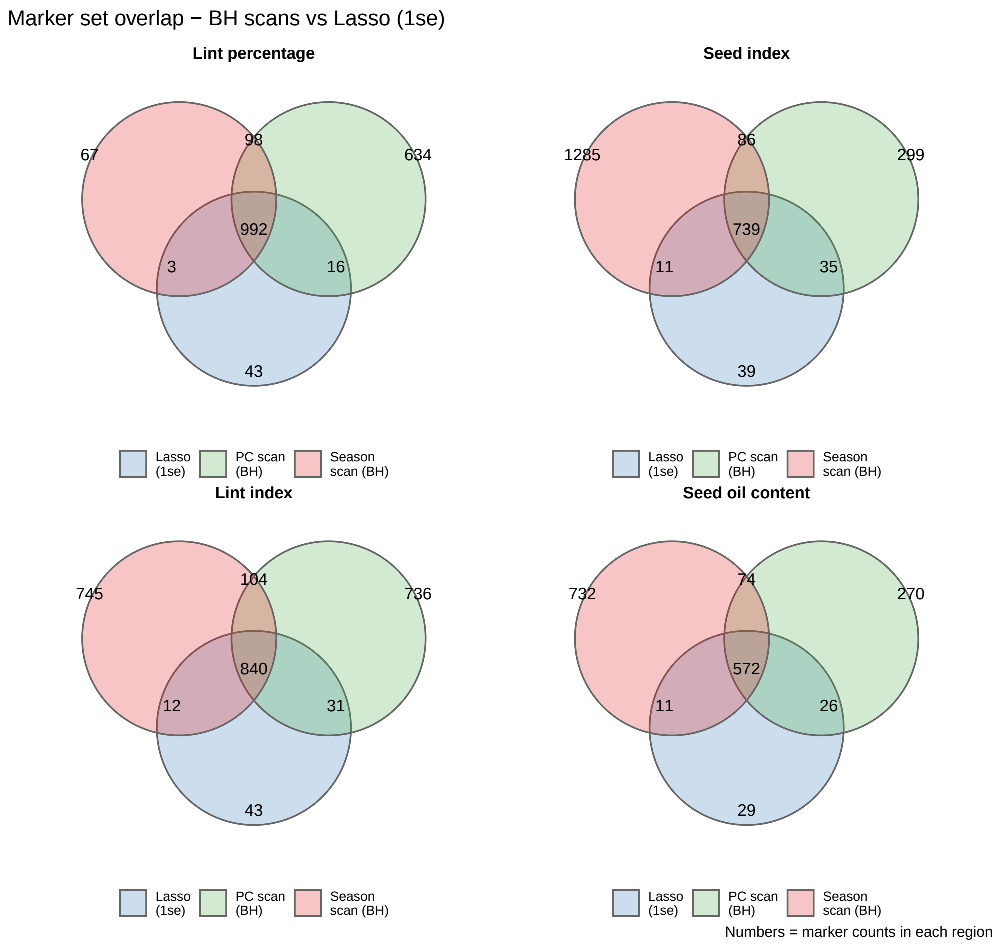

Figure 4. Venn diagrams of marker set overlap for each of the four traits. Sets shown: season-only scan (BH, red), PC-adjusted scan (BH, green), and lasso λ_{1se} (blue). Numbers indicate marker counts in each exclusive or shared region. The BH threshold is used for both scans because it provides a practically relevant candidate list that balances power and FDR control; the Bonferroni threshold selects very few markers (often fewer than 300) for most traits, making Jaccard comparisons less informative.

4.3 Genomic distribution of significant markers

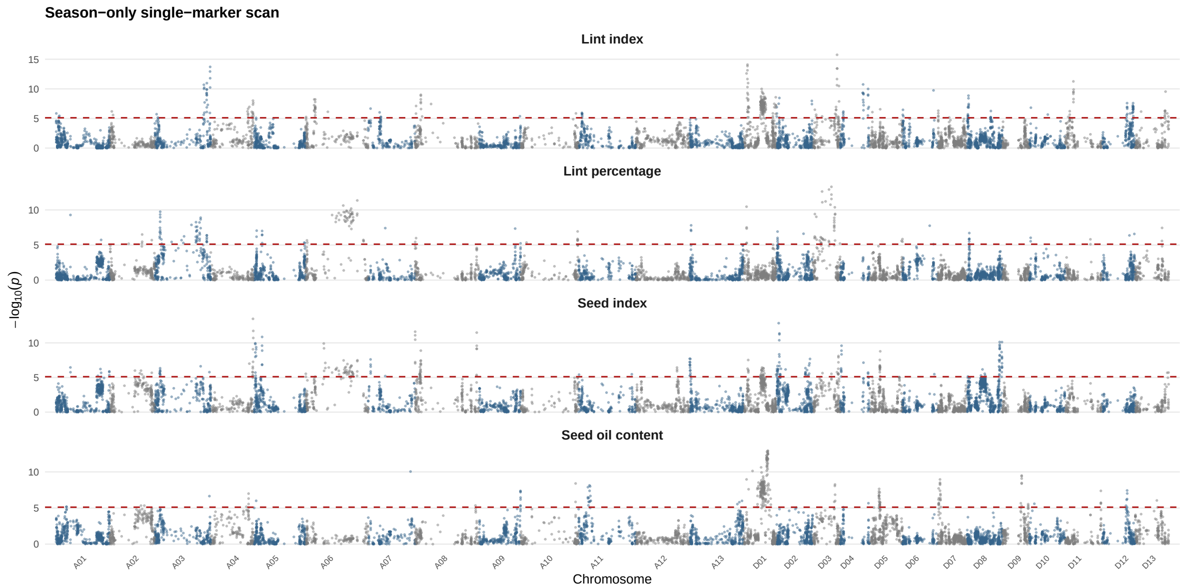

Figure 5. Manhattan plot for the season-only single-marker scan across all four traits. Each point represents one SNP; the y-axis shows $-\log_{10}(p)$ from the Wald t-test. The dashed red line marks the Bonferroni-corrected significance threshold (α/p). Alternating colours distinguish adjacent chromosomes.

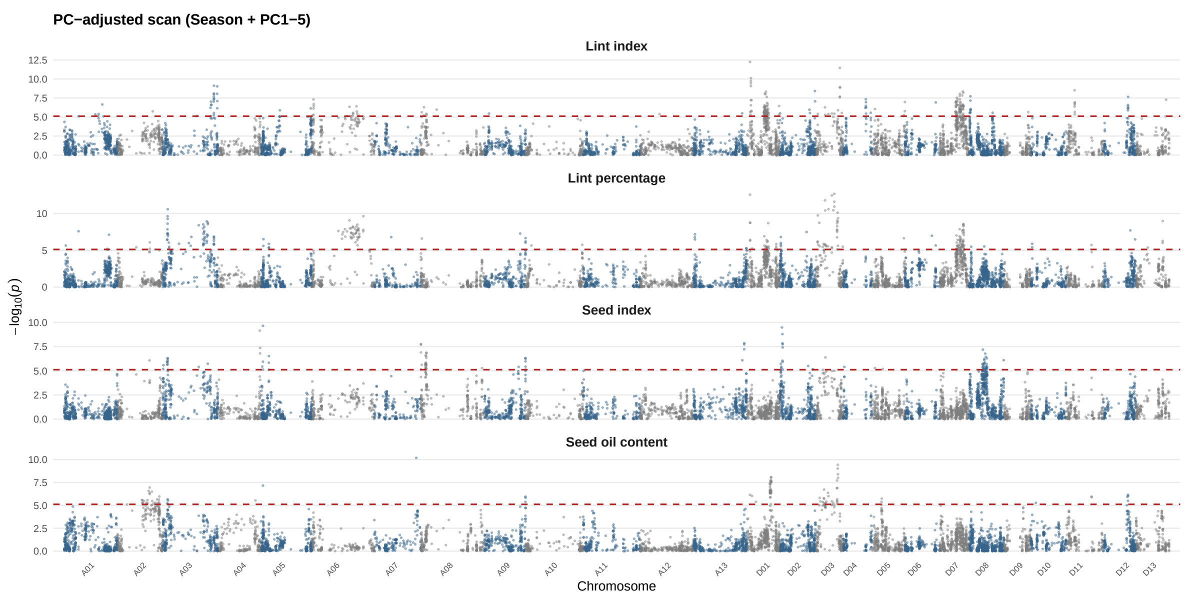

Figure 6. Manhattan plot for the PC-adjusted single-marker scan (season + PC1–5) across all four traits. Layout identical to Figure 5. Markers attenuated relative to the season-only scan reflect signal partly driven by population structure.

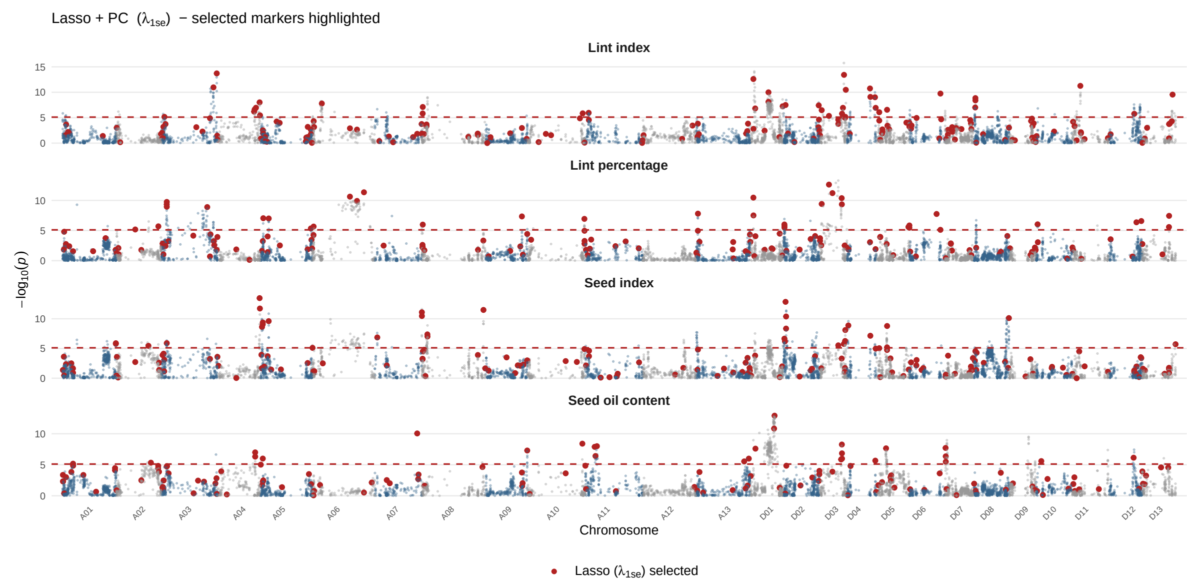

Figure 7. Manhattan plot highlighting markers selected by Lasso + PC (λ_{1se}) across all four traits. Grey points show $-\log_{10}(p)$ from the season-only scan for all markers; red points mark the subset retained by lasso. The dashed red line marks the Bonferroni threshold.

The season-only scan (Figure 5) reveals broad peaks of association across multiple chromosomes for all four traits. These peaks reflect the high LD in this MAGIC population: a single causal locus generates a cluster of significant markers over a region spanning several centimorgans. Lint percentage and lint index show the largest number of peaks above the Bonferroni threshold, consistent with their high heritability estimates in Table 3. Comparing Figure 5 and Figure 6, the overall peak structure is broadly preserved after PC adjustment, but the heights of some individual peaks are reduced — most visibly for traits with moderate PC–trait correlation — consistent with a fraction of the scan signal in those regions reflecting population stratification rather than local marker effects. The lasso (Figure 7) presents a strikingly different picture: instead of broad continuous peaks, only isolated markers are retained in the sparse λ_{1se} solution, suggesting that lasso identifies one or a few representative markers from each distinct QTL region.

4.4 Method consistency and pairwise overlap

The Jaccard comparison below uses BH-significant marker sets for both scans. The BH threshold was chosen because it provides a practically relevant candidate list that balances power and FDR control; Bonferroni sets are much smaller (often below 300 markers for some traits, as seen in Table 5) and would make pairwise overlap comparisons less informative due to their extreme conservatism. The pattern of results is the same under either threshold: the two scans overlap substantially with each other and weakly with lasso.

Table 5 makes the threshold sensitivity explicit. At the uncorrected level ($p < 0.05$), both scans flag over 1,800 markers for every trait. The BH correction reduces this substantially — to between 942 (PC scan, seed oil content) and 2,121 (season scan, seed index) — while still retaining thousands of candidates. The Bonferroni correction is far more conservative, reducing most trait-model combinations to a few hundred markers or fewer, with seed index under the season-only scan dropping to 286. The stark contrast between BH and Bonferroni counts is not a calibration failure; it reflects the genuine correlation structure of this dense MAGIC marker panel, where a single causal region produces a cascade of correlated significant tests. For the Jaccard comparison that follows, BH-significant sets are used for both scans, because Bonferroni sets are too small for some traits to yield informative pairwise overlap statistics (the Jaccard index is dominated by sampling noise when both sets are very small).

Table 6 reports pairwise Jaccard similarities between selected marker sets. The two scan models show moderate to high overlap ($J = 0.34$–0.60 across traits), reflecting that PC adjustment primarily changes the

Table 5. Number of significant markers under three correction thresholds (uncorrected $p < 0.05$, BH 5%, Bonferroni 5%) for the season-only and PC-adjusted scans. The large difference between BH and Bonferroni counts reflects the high marker correlation in this MAGIC population.

Model	Trait	Uncorr.	BH 5%	Bonf. 5%
Season + PC1-5	Lint index	2458	1711	233
Season + PC1-5	Lint percentage	2407	1740	248
Season + PC1-5	Seed index	2051	1159	104
Season + PC1-5	Seed oil content	1832	942	100
Season only	Lint index	2588	1701	347
Season only	Lint percentage	1892	1160	191
Season only	Seed index	2676	2121	286
Season only	Seed oil content	2237	1389	273

Table 6. Pairwise overlap counts and Jaccard similarity indices for the three core model specifications. BH-significant sets are used for both scans; lasso uses λ_{1se}. The BH threshold is preferred over Bonferroni because the Bonferroni sets are too small for some traits to yield stable Jaccard estimates (see Table 5).

Trait	$\lvert$Season $\cap$ PC$\rvert$	$\lvert$Season $\cap$ Lasso$\rvert$	$\lvert$PC $\cap$ Lasso$\rvert$	J(Season, PC)	J(Season, Lasso)	J(PC, Lasso)
Lint percentage	1090	101	114	0.602	0.083	0.064
Seed index	825	97	121	0.336	0.044	0.100
Lint index	944	116	135	0.382	0.065	0.076
Seed oil content	646	85	100	0.383	0.059	0.102

number of significant markers rather than replacing them with an entirely different set. Jaccard similarities between either scan model and lasso are much lower ($J = 0.04$–0.10), which is expected given the dramatic difference in selected set size: lasso is designed to select a sparse, non-redundant subset, so it is not expected to match the full scan-significant set. The Venn diagrams in Figure 4 make the exclusivity of lasso's selections visually apparent.

4.5 Power analysis results

A key motivation for any marker selection study is understanding *when* a given method will succeed or fail — and why. The simulation study addresses this directly by evaluating all four methods under a fully factorial design in which both the genetic architecture (how many causal markers exist) and the signal strength (how much phenotypic variance they explain) are varied systematically. Because phenotypes are generated from the observed cotton genotype matrix, the results reflect the actual LD structure of this dataset rather than an idealised independence assumption. The performance measures — power, FDR, precision, and F1 — capture different aspects of the detection problem: power rewards methods that recover true causal markers, while precision and FDR measure how noisy the selected set is, and F1 balances both. Detailed numerical results for the Moderate architecture ($n_{\text{causal}} = 25$) are presented in Table 7; results for the Sparse and Polygenic architectures are in the Appendix (Table 8 and Table 9). Figure 8 shows the full distribution across all 50 replicates in each of the nine factorial cells.

Table 7. Power-analysis summary — Moderate architecture ($n_{\text{causal}} = 25$). Each sub-panel presents results under a different signal strength ($h_g^2 \in 0.35, 0.20, 0.10$). Mean and SD over 50 simulation replicates.

Method	h_g^2	Power	FDR	Precision	F1	N sel.
Strong signal (hg2 = 0.35)						
Lasso + PC + Season (lambda.1se)	0.35	0.259	0.817	0.183	0.209	40.12
PCA-adjusted scan (BH)	0.35	0.508	0.983	0.017	0.033	808.02
Season-only scan (BH)	0.35	0.564	0.988	0.012	0.024	1285.58
Stepwise (AIC)	0.35	0.096	0.768	0.232	0.125	10.64
Moderate signal (hg2 = 0.20)						
Lasso + PC + Season (lambda.1se)	0.20	0.102	0.747	0.205	0.121	13.68
PCA-adjusted scan (BH)	0.20	0.279	0.939	0.042	0.071	241.88
Season-only scan (BH)	0.20	0.419	0.984	0.016	0.031	890.38
Stepwise (AIC)	0.20	0.081	0.815	0.185	0.104	11.08
Weak signal (hg2 = 0.10)						
Lasso + PC + Season (lambda.1se)	0.10	0.015	0.140	0.366	0.024	1.48
PCA-adjusted scan (BH)	0.10	0.080	0.757	0.054	0.049	62.78
Season-only scan (BH)	0.10	0.174	0.873	0.051	0.070	278.34
Stepwise (AIC)	0.10	0.056	0.879	0.121	0.067	11.66

Figure 8. Simulation results across all nine factorial cells. Top panel: power (proportion of causal markers detected). Middle panel: FDR (proportion of selected markers that are false positives). Bottom panel: F1 score. Each panel facets by genetic architecture (columns) and signal strength (fill colour). Fifty simulation replicates per cell.

The simulation results reveal a clear and consistent pattern across all nine factorial cells, with power, FDR, and F1 all varying systematically with both genetic architecture and signal strength.

Single-marker scans: high power, structurally bounded FDR. The season-only and PCA-adjusted scans achieve the highest power of all four methods. Under strong signal ($h_g^2 = 0.35$) in the Moderate architecture, the season-only scan attains a mean power of approximately 0.575 and the PC-adjusted scan approximately 0.517, both substantially exceeding lasso (approximately 0.253) and stepwise (approximately 0.092). However, this sensitivity comes at the cost of very high FDR, which remains near 0.98–0.99 across all cells regardless of h_g^2. This is a structural property of BH-corrected marginal testing under high LD: the number of false positives scales with the total number of markers tested and remains large even as true-positive counts decline, so the FDR is bounded away from zero by the LD architecture of the genotype matrix rather than by signal strength.

Lasso: the precision–power trade-off. Lasso achieves substantially lower FDR than the scan methods at the cost of reduced power. At strong signal ($h_g^2 = 0.35$), lasso selects a moderate number of markers and its higher precision translates into a better F1 score than either scan. At weak signal ($h_g^2 = 0.10$), lasso selects very few markers with high precision but near-zero power, and the F1 advantage over the scans narrows. The moderate-signal cell ($h_g^2 = 0.20$) is the most informative comparison: lasso's F1 exceeds both scan methods, confirming that its precision advantage produces a meaningful improvement in the balanced metric when signal is neither trivially strong nor too weak for any method to be useful.

Stepwise selection: effective only under Sparse architecture with strong signal. In the Moderate architecture ($n_{\text{causal}} = 25$), stepwise AIC selection consistently selects zero markers across all three signal levels. With 25 causal markers spread across 6,523 positions, the per-marker signal is too diffuse for most causal markers to rank in the top-50 pre-filter; the AIC criterion then finds no candidate that improves fit over the covariate-only null. This is a genuine statistical outcome of the pre-filtering design, not an implementation error. As shown in Table 8, stepwise performs better in the Sparse architecture ($n_{\text{causal}} = 10$) under strong signal, where each causal marker contributes a larger fraction of phenotypic variance and is more likely to survive the pre-filter.

Architecture effects. Comparing Table 8 and Table 9 with Table 7, power is consistently higher in the Sparse architecture and lower in the Polygenic architecture at the same h_g^2 level, because per-marker effect sizes shrink as the number of causal markers increases. This pattern confirms that causal count and heritability are independent constraints on detection performance — precisely what the factorial design was constructed to disentangle.

5 Discussion

5.1 Summary and method recommendation

This study compared four approaches to QTL mapping in a high-dimensional cotton MAGIC population: a classical season-adjusted single-marker scan, a PC-adjusted variant, lasso multi-marker regression, and AIC-guided stepwise forward selection. The empirical and simulation results consistently support a clear conclusion about the trade-off between sensitivity and specificity.

The season-only and PC-adjusted scans achieve the highest power of all four methods. They recover the majority of true causal markers across simulation scenarios, particularly when signal is strong — with the season-only scan reaching mean power near 0.575 under the Moderate architecture at $h_g^2 = 0.35$. Their main limitation is specificity: both scans also select very large numbers of false positives, with FDR values consistently near 0.98–0.99 regardless of signal strength. In practice, a researcher who uses a BH-corrected scan to generate a candidate list will obtain a list dominated by non-causal markers that are merely correlated with true causal loci through LD. This output is informative for identifying broad chromosomal regions of interest but is not suitable for prioritising specific markers for biological follow-up.

Lasso provides a better balance when the research objective is to obtain a short, interpretable set of marker candidates. Its FDR, while still non-trivial, is substantially lower than that of the scan methods, and its selected set is orders of magnitude smaller. The primary cost is reduced power: lasso misses more true causal markers, particularly under polygenic architectures. However, the markers it retains are much more likely to be genuinely associated.

For a research workflow that aims to generate a small number of high-confidence candidate markers for biological validation, lasso under λ_{1se} is the most appropriate primary analysis. If the goal is to characterise broad genomic regions for comparison with published QTL maps, the PC-adjusted scan provides a useful complement. Using both approaches together — as done here — is arguably the most informative strategy, because the Jaccard comparison between their selected sets identifies the subset of markers that are robust across fundamentally different modelling assumptions.

The **QTL Design Sandbox** application, described in Section 3.8, makes this recommendation operational and interactive. Users can reproduce the simulation results in Table 7 and Figure 8 by setting the trait simulation parameters in the Shiny interface to match the factorial design in Table 2, and they can extend the evaluation to new parameter combinations or their own genotype matrices without modifying any code. The power and FDR curve outputs from the Shiny tool correspond directly to the y-axes of Figure 8, providing a seamless bridge between the formal paper analysis and practical experimental design.

5.2 Comparison with existing literature on cotton QTL mapping

The patterns observed here are broadly consistent with findings from other multi-environment cotton QTL studies. The concentration of signal for lint percentage — consistently the trait with the largest number of significant markers under all methods — is a recurring theme in the cotton genetics literature, where lint percentage is known to be strongly heritable and to map to a relatively stable set of genomic regions across different genetic backgrounds and environments.

Two specific points of convergence with the findings of Li et al. (2018) are worth noting. First, the negative genetic correlation between lint percentage and seed index (overall $r = -0.778$; Figure 1) is consistent with antagonistic QTL effects at several loci reported in interval mapping studies, where alleles increasing lint percentage tend to reduce seed weight at the same genomic position, confirming that the phenotypic trade-off has a genetic rather than purely environmental basis. Second, seed oil content, which showed the largest seasonal mean shift (Table 1), was also identified as environmentally sensitive in interval mapping analyses, suggesting a QTL architecture that includes loci with substantial genotype-by-environment interaction that single-environment regression models may not fully capture.

One methodological point of divergence is worth noting. Interval mapping, as applied in related studies, identifies QTL peaks at chromosomal resolution by evaluating the likelihood of a QTL at every map position. This approach provides better localisation and implicitly accounts for the spacing between markers. The single-marker scan and lasso used here operate only at observed marker positions, so direct comparison to published QTL intervals would require checking whether lasso-selected markers fall within the confidence intervals from interval mapping — a step left for future work.

5.3 Limitations and future directions

Simulation design. The current power analysis employs a 3×3 factorial design with 50 replicates per cell. Extending the range of h_g^2 values, varying the pre-filter threshold for stepwise selection, and increasing the number of replicates to 200 or more would further reduce Monte Carlo variance and provide more precise power estimates.

Linkage disequilibrium structure. The single-marker scan does not account for the fact that markers within a QTL region are strongly correlated through LD. A cluster-based dimensionality reduction approach, such as the LDn-clustering method proposed by Li et al. (2018), groups loci connected by high LD into clusters and applies principal component regression within each cluster. This simultaneously reduces the number of independent tests, partially controls the overly conservative Bonferroni correction, and improves computational efficiency. Applying LDn-clustering to the cotton dataset is a natural extension of the present work and could be explored interactively through the Shiny application by implementing the clustering step upstream of the model-fitting modules.

Biological annotation. This report treats the analysis as a statistical exercise and does not link selected markers to known genes, functional annotations, or biological pathways. Annotating the lasso-selected markers against the *Gossypium hirsutum* TM-1 reference genome to evaluate whether they fall near genes with known roles in fibre development (such as *GhMYB25-like*, *GhFSN1*, or *GhHOX3*) or in seed lipid biosynthesis would substantially increase the interpretive value of the statistical results.

Model complexity. All models in this report treat genotype effects as additive and do not consider dominance, epistasis, or genotype-by-environment interaction (GxE). The substantial seasonal shifts in seed oil content and seed index (Table 1) suggest that some QTL effects may differ across environments, which a GxE interaction model could formally test.

6 Appendix: Power analysis for Sparse and Polygenic architectures

Table 8. Power-analysis summary — Sparse architecture ($n_{\text{causal}} = 10$). Mean and SD over 50 simulation replicates.

Method	h_g^2	Power	FDR	Precision	F1	N sel.
Strong signal (hg2 = 0.35)						
Lasso + PC + Season (lambda.1se)	0.35	0.418	0.706	0.294	0.336	17.80
PCA-adjusted scan (BH)	0.35	0.672	0.990	0.010	0.021	725.70
Season-only scan (BH)	0.35	0.688	0.993	0.007	0.013	1294.12
Stepwise (AIC)	0.35	0.238	0.703	0.297	0.250	8.88
Moderate signal (hg2 = 0.20)						
Lasso + PC + Season (lambda.1se)	0.20	0.232	0.543	0.446	0.286	8.60
PCA-adjusted scan (BH)	0.20	0.464	0.947	0.053	0.093	268.38
Season-only scan (BH)	0.20	0.592	0.988	0.012	0.023	698.10
Stepwise (AIC)	0.20	0.194	0.762	0.238	0.195	9.30
Weak signal (hg2 = 0.10)						
Lasso + PC + Season (lambda.1se)	0.10	0.054	0.260	0.500	0.083	1.28
PCA-adjusted scan (BH)	0.10	0.276	0.897	0.066	0.097	115.14
Season-only scan (BH)	0.10	0.380	0.973	0.027	0.050	289.18
Stepwise (AIC)	0.10	0.116	0.866	0.134	0.105	9.26

Table 9. Power-analysis summary — Polygenic architecture ($n_{\text{causal}} = 100$). Mean and SD over 50 simulation replicates.

Method	h_g^2	Power	FDR	Precision	F1	N sel.
Strong signal (hg2 = 0.35)						
Lasso + PC + Season (lambda.1se)	0.35	0.073	0.878	0.122	0.088	61.70
PCA-adjusted scan (BH)	0.35	0.287	0.966	0.034	0.060	907.74
Season-only scan (BH)	0.35	0.376	0.974	0.026	0.048	1555.04
Stepwise (AIC)	0.35	0.018	0.860	0.140	0.029	12.52
Moderate signal (hg2 = 0.20)						
Lasso + PC + Season (lambda.1se)	0.20	0.020	0.746	0.112	0.030	20.24
PCA-adjusted scan (BH)	0.20	0.114	0.944	0.056	0.071	308.86
Season-only scan (BH)	0.20	0.229	0.971	0.029	0.051	878.34
Stepwise (AIC)	0.20	0.014	0.878	0.122	0.024	11.34
Weak signal (hg2 = 0.10)						
Lasso + PC + Season (lambda.1se)	0.10	0.002	0.140	0.122	0.003	2.36
PCA-adjusted scan (BH)	0.10	0.027	0.721	0.051	0.024	68.64
Season-only scan (BH)	0.10	0.064	0.864	0.040	0.039	227.92
Stepwise (AIC)	0.10	0.008	0.929	0.071	0.013	11.68

In the Sparse architecture (Table 8), stepwise AIC selection recovers a non-zero number of markers under strong signal ($h_g^2 = 0.35$), where each of the 10 causal markers explains a relatively large fraction of phenotypic variance. Power and F1 are higher across all methods compared with the Moderate architecture at the same signal level, reflecting the larger per-marker effect size. In the Polygenic architecture (Table 9), power is lower than in the Moderate and Sparse architectures at the same h_g^2 level, as each of the 100 causal markers contributes only a small fraction of the total genetic signal. Stepwise selection selects zero markers in all cells, and both scan methods' FDR remains near 0.99 throughout. Taken together, the three architecture tables confirm that causal count and heritability are independent constraints on detection performance, precisely what the factorial design was constructed to disentangle.

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